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GitHub Link	**https://github.com/YOGAMADHA/MLA0405-DEEP-LEARNING**

SIMATS ENGINEERING

CO2 - ASSESSMENT TOOL 2

Scenario-Based Questions

COURSE NAME: Deep Learning **COURSE CODE:MLA0405**
Faculty : Dr. Arul Raj A.M **Slot : D**

COURSE OUTCOME COVERED

CO 2: Students will be able to apply supervised learning algorithms such as linear regression, classification models, decision trees, neural networks, support vector machines, and ensemble methods to solve real-world prediction and classification problems. They will gain the ability to select appropriate models, train them using datasets, and evaluate their performance. Students will also understand the strengths and limitations of different supervised learning approaches.(BTL-4)

CO Weightage : 15%

Assessment Tool Weightage: 40%

Duration: 90 minutes

Marks: 25

Code of Conduct:

I **Yoga Madha A-192425058** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Yogamadha

Signature of the student: **Name of the student: Yoga Madha A**

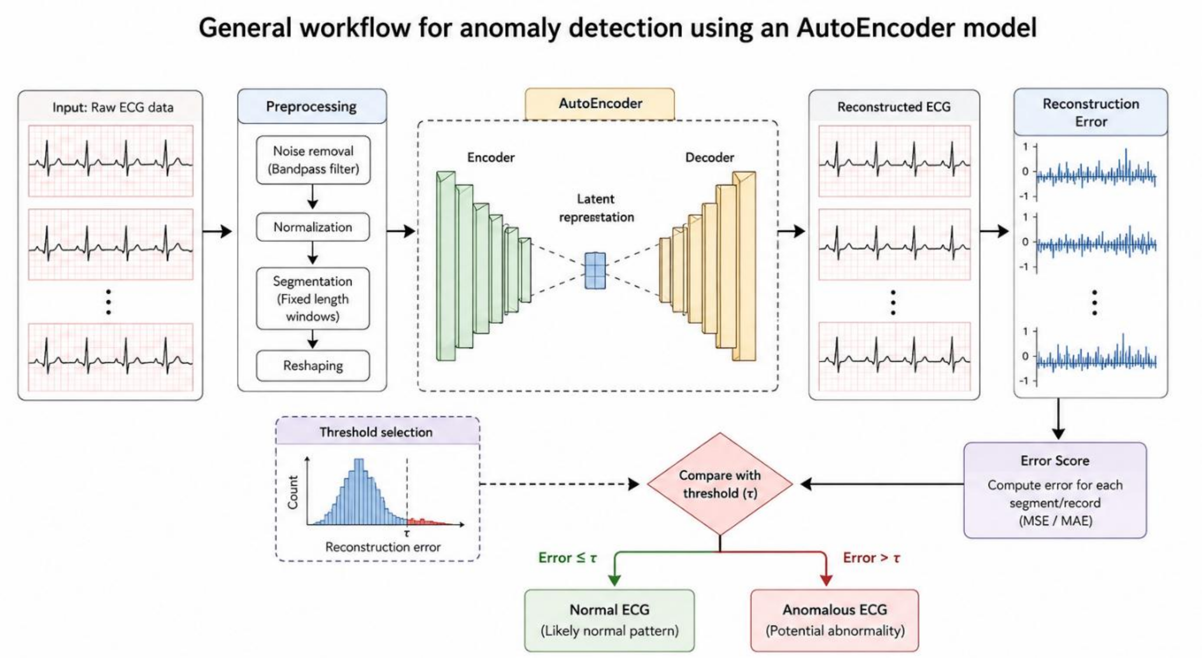
Scenario-Based Questions

CO-AT2

QUESTIONS: 05 **Total Marks:25**

1. **Unsupervised Training of Neural Networks, Auto Encoders**
   A hospital wants to identify abnormal ECG patterns from thousands of unlabeled records. Which deep learning technique from the syllabus would you recommend? Justify your choice and explain the workflow.
   Answer:
   For identifying **abnormal ECG patterns from thousands of unlabeled ECG records**, I would recommend an **Autoencoder** because it is an **unsupervised deep learning technique** that can learn patterns without requiring labeled data.

   Workflow



1. **Collect ECG data:** Gather thousands of ECG records without labels.
2. **Preprocess:** Remove noise, normalize the signals, and divide them into suitable segments.
3. **Train the Autoencoder:** The **encoder** compresses the ECG signal into a smaller latent representation.
4. **Reconstruct the ECG:** The **decoder** reconstructs the original ECG from the compressed representation.
5. **Calculate reconstruction error:** Compare the original ECG with the reconstructed ECG.
6. **Detect abnormalities:** Normal ECG patterns are reconstructed with **low error**, while unusual or abnormal patterns generally produce **higher reconstruction error**.
7. **Set a threshold:** If the reconstruction error exceeds a selected threshold, the ECG can be flagged for further medical examination.

Justification

An **Autoencoder is suitable because the ECG records are unlabeled**. It learns the underlying patterns of the available data without requiring a doctor to manually label every ECG as normal or abnormal. This makes it useful for large-scale **anomaly detection** and can help healthcare professionals identify ECG records that need further investigation.

2. **Auto Encoders, Representation Learning**

A retail company wants to reduce the dimensionality of customer purchase data before performing customer segmentation. Explain how an autoencoder can be used to solve this problem.

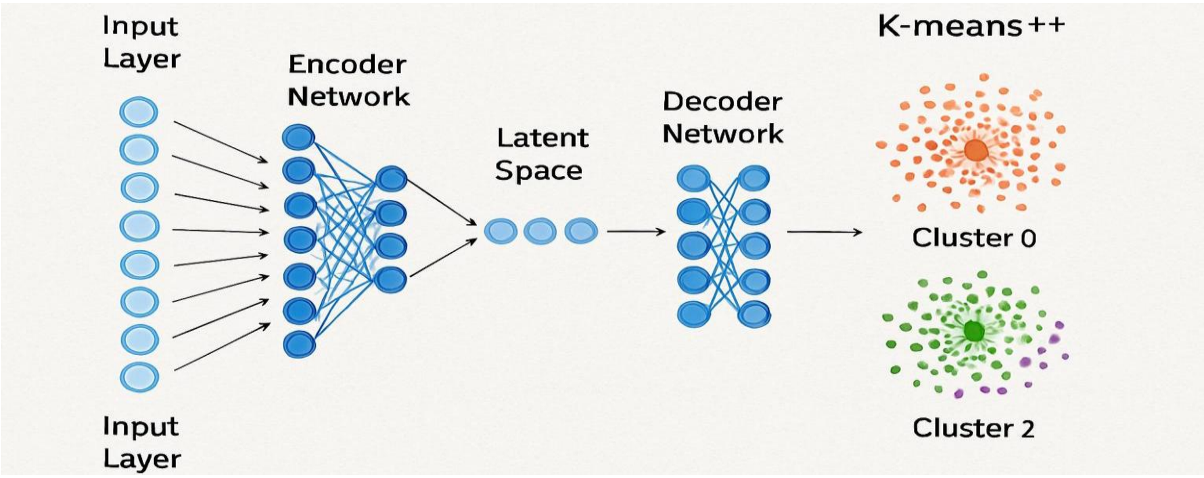
Answer:

An **Autoencoder** is an unsupervised neural network that learns a compressed representation of the input data. It consists of two parts:

- **Encoder** – Compresses the customer purchase data into a lower-dimensional representation.
- **Decoder** – Reconstructs the original data from the compressed representation.

The compressed representation (latent features) retains the most important information while removing redundant features. These latent features can then be used for customer segmentation.

Workflow



Steps

1. Data Collection:

Collect customer purchase data such as product purchases, spending amount, purchase frequency, and transaction history.

2. Preprocessing:

Clean the data, handle missing values, and normalize the features.

3. Encoder:
The encoder compresses the high-dimensional customer data into a smaller set of important features.
4. Latent Representation:
The compressed data contains the essential information needed to represent customer behavior.
5. Decoder:
The decoder reconstructs the original data from the compressed representation.
6. Customer Segmentation:
The compressed features are used as input to clustering algorithms (such as K-Means) to group customers with similar purchasing behavior.

Advantages

- Reduces data dimensionality.
- Removes redundant and unnecessary features.
- Preserves important customer information.
- Improves clustering performance and reduces computation time.

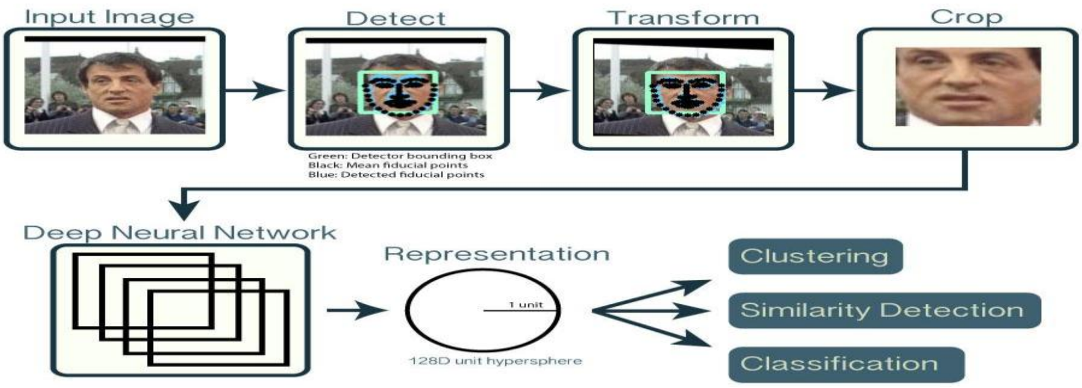
Conclusion

An **Autoencoder** is an effective technique for reducing the dimensionality of customer purchase data. It learns a compact representation of the data while preserving essential information. The reduced features can then be used to perform accurate and efficient customer segmentation.

3. **Width and Depth of Neural Networks, Activation Functions**
A facial recognition system performs poorly when trained using a shallow neural network. Recommend architectural changes involving network depth and activation functions.
Answer:
A **deep neural network** should be used instead of a shallow network. Increasing the number of hidden layers allows the model to learn complex facial features such as edges, eyes, nose, and facial expressions.
For activation functions:

- **ReLU (Rectified Linear Unit)** should be used in the hidden layers because it speeds up training and reduces the vanishing gradient problem.
- **Softmax** should be used in the output layer for multi-person face classification.

Workflow



Steps

1. Input Image:
Capture and preprocess the face image.

2. Deep Neural Network:
Use multiple hidden layers to learn complex facial features.

3. ReLU Activation:
Apply ReLU in hidden layers to improve learning speed and overcome vanishing gradients.

4. Feature Extraction:
The deep network extracts important facial characteristics.

5. Softmax Output:
The output layer predicts the most likely person's identity.

Advantages

- Learns complex facial features.
- Improves recognition accuracy.
- Faster training with ReLU.
- Softmax provides accurate multi-class classification.

Conclusion

A **deep neural network** with **multiple hidden layers** and **ReLU activation** in the hidden layers is recommended for facial recognition. Using **Softmax** in the output layer improves classification accuracy, resulting in better recognition performance than a shallow neural network.

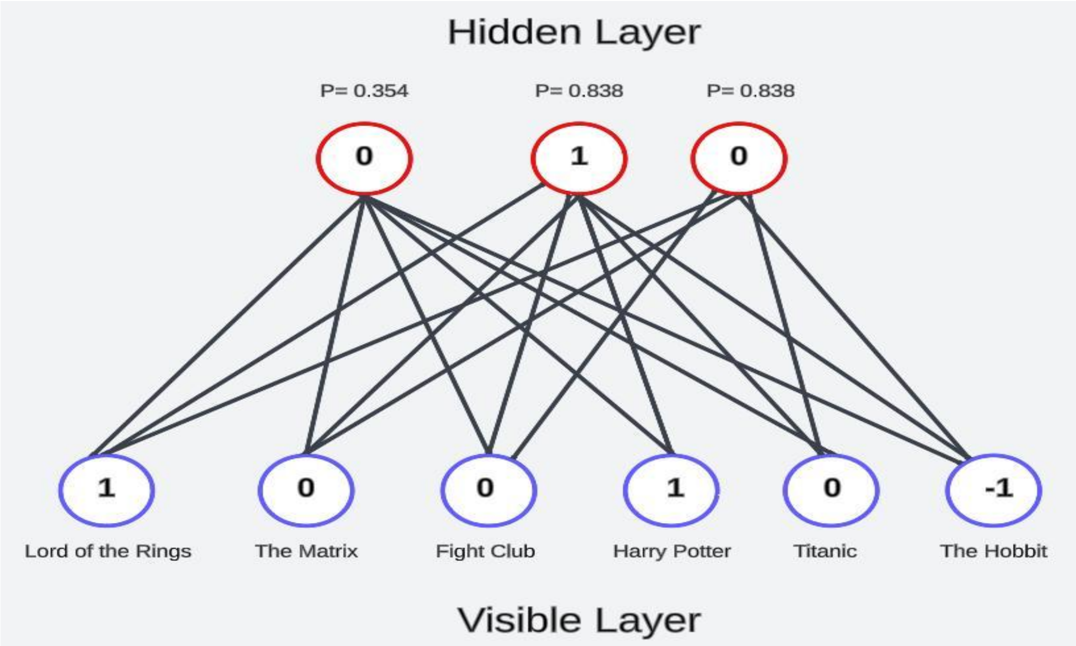
4. Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks

A streaming service wants to recommend movies to users based on their viewing history without using labeled data. Explain how RBMs can be applied to build the recommendation engine.

Answer:

A **Restricted Boltzmann Machine (RBM)** is an unsupervised neural network that learns hidden patterns from user–movie interactions. It consists of a **Visible Layer** (users' movie ratings/viewing history) and a **Hidden Layer** (latent features such as movie preferences).
After training, the RBM predicts which movies a user is likely to watch or enjoy, even if they have never seen them before.

Workflow



Steps
1. Data Collection:
Collect users' viewing history or movie ratings.
2. Preprocessing:
Convert the viewing history into a user–movie matrix.
3. Visible Layer:
The visible layer stores information about movies watched or rated by users.
4. Hidden Layer:
The hidden layer learns hidden preferences, such as favorite genres, actors, or movie styles.
5. Recommendation:
The RBM predicts ratings for unseen movies based on the learned preferences.
6. Movie Suggestion:
Movies with the highest predicted ratings are recommended to the user.

Advantages

- Works with **unlabeled data**.
- Learns hidden user preferences automatically.
- Handles large movie datasets efficiently.
- Provides personalized movie recommendations.

Conclusion

A **Restricted Boltzmann Machine (RBM)** is an effective technique for building a movie recommendation system using **unsupervised learning**. It learns hidden patterns from users' viewing history and predicts movies they are likely to enjoy, enabling personalized recommendations without requiring labeled data.

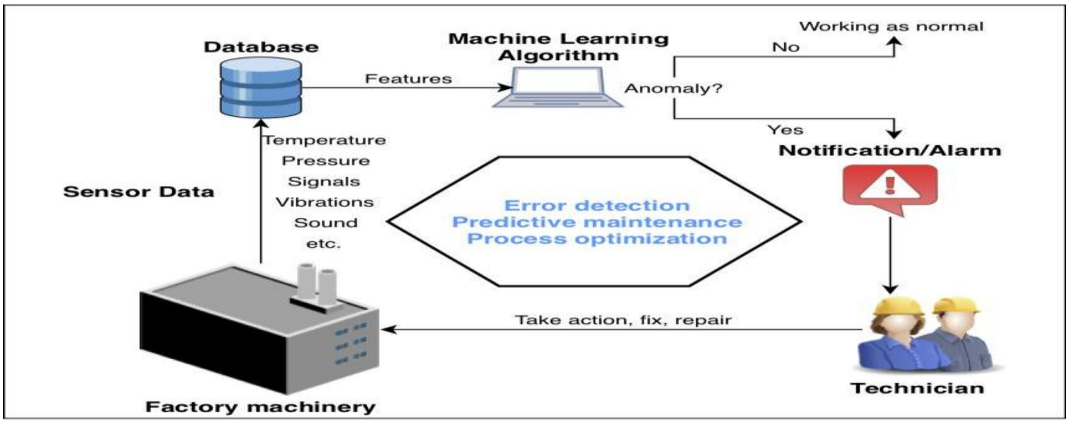
5. **Deep Learning Applications, Representation Learning**

A manufacturing company wants to detect defective products from sensor readings collected every second. Design a deep learning solution using concepts from representation learning and neural networks

Answer:

Representation Learning enables the neural network to automatically learn useful features from the sensor data instead of relying on manually designed features. The learned features are then used to classify products as **Normal** or **Defective**.

Workflow



Steps

1. Data Collection:

Collect sensor readings such as temperature, pressure, vibration, and speed every second.

2. Preprocessing:

Clean the data, remove noise, and normalize the sensor values.

3. Representation Learning:

The deep neural network automatically learns important features from the sensor data.

4. Neural Network Classification:

The learned features are passed through hidden layers to classify the product.

5. Defect Detection:

The output layer predicts whether the product is **Normal** or **Defective**.

Advantages

- Learns features automatically from sensor data.
- Detects defects with high accuracy.
- Handles large volumes of real-time sensor data.
- Reduces manual inspection and improves quality control.

Conclusion

A **deep neural network using representation learning** is an effective solution for detecting defective products from continuous sensor readings. It automatically learns meaningful features and accurately classifies products as **normal** or **defective**, improving manufacturing quality and efficiency.

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand